Project 1: Constant returns to scale in production

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1 Introduction

Constant returns to scale in production is a very common assumption within economic models, often motivated by the replication argument. In this project, we test this assumption utilizing a panel dataset of French manufacturing firms spanning 12 years. We estimate the parameters in a Cobb-Douglas production function using a first-difference model and find no evidence in favor of constant returns to scale.

2 Empirical setting and Data

In this project we seek to empirically investigate whether firms exhibit constant returns to scale. We consider a Cobb-Douglas production function

$$F(K, L) = AK^{\beta_K} L^{\beta_L} \tag{1}$$

, which takes labor (L) and capital (K) as input factors. A represents total factor productivity (TFP). For a Cobb-Douglas production function, constant returns to scale implies that $\beta_K + \beta_L = 1$. To investigate this, we use a balanced panel dataset covering 441 French manufacturing firms from 1968-1979. The dataset contains log of deflated sales, log of employment, and log of adjusted capital stock. The variables are demeaned by subtracting the cross-sectional mean for each year.

3 Methodology

i.i.d assumption is implicit
But we need it for a LLN (and CLT) to apply

We take the logarithm of equation 1 to obtain the following estimable equation

$$y_{i,t} = \mathbf{x}_{i,t}\beta + v_{i,t}, \text{ with } v_{i,t} = c_i + u_{i,t} \tag{2}$$

, where $y_{i,t}$ is the log of deflated sales for firm i in year t, $\beta = (\beta_K \ \beta_L)'$, and $\mathbf{x}_{i,t} = (k_{i,t} \ \ell_{i,t})$. $k_{i,t}$ is the log of adjusted capital stock and $\ell_{i,t}$ is the log of employment. $v_{i,t}$ represents the log of the total factor productivity ($ln(A_{i,t})$) of each firm, which we cannot observe. As described in section 2, all variables are demeaned, such that a common intercept by construction would equal zero and therefore estimating an intercept is redundant. Figure 1 provides visual evidence of clustering of observations by firm, suggesting that firms differ systematically in their productivity. We therefore decompose TFP into a firm-specific, time-invariant component (c_i) and a time-varying productivity shock ($u_{i,t}$).

A testable implication of constant return to scale is that $\beta_K + \beta_L = 1$. For our test to

be meaningful, we need consistent estimators of β_K and β_L and the ability to gauge their uncertainty. The simplest approach is to employ pooled OLS (POLS).
For $\hat{\beta}_{POLS} = (\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbf{x}'_{i,t}\mathbf{x}_{i,t})^{-1}(\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbf{x}'_{i,t}y_{i,t})$ to be consistent, we need an orthogonality condition and a rank condition, i.e. POLS.1: $E[\mathbf{x}'_{i,t}v_{i,t}] = 0 \ \forall \ t$ and POLS.2: $rank(\sum_{t=1}^{T}E[\mathbf{x}'_{i,t}\mathbf{x}_{i,t}]) = 2$. From POLS.1, we require $E[\mathbf{x}'_{i,t}u_{i,t}] = 0 \ \forall \ t$, which might seem reasonable in light of the classic economic assumption that firms cannot adjust their factors of production in the very short run and would therefore be unable to react immediately to a productivity shock. However, a second requirement is that $E[\mathbf{x}'_{i,t}c_i] = 0 \ \forall \ t$, which may be more problematic. Figure 1 does not provide any clear visual evidence of whether the time-invariant component of firm productivity is correlated with the quantity of the input factors. However, if e.g. managerial talent increases the marginal product of input factors, we would expect firms with high managerial talent to use more capital and labor leading $\hat{\beta}_{POLS}$ to be inconsistent.

To alleviate this concern, we exploit the panel structure of the data to account for any arbitrary dependence between the time-invariant component of firm productivity (c_i) and the input factors ($k_{i,t}, \ell_{i,t}$). We can remove the time-invariant component of firm productivity by applying either a within transformation (FE) or taking first differences (FD). In our empirical analysis we examine the validity of both a Fixed Effects- and First Difference model.

For the fixed-effects model, we within transform equation 2 which yields the following equation

$$\ddot{y}_{i,t} = \ddot{\mathbf{x}}_{i,t}\beta + \ddot{u}_{i,t} \quad \forall \ t = 1, 2, .., 12 \tag{3}$$

, where the dots represent the demeaned variables, e.g. $\ddot{y}_{i,t} = y_{i,t} - \frac{1}{12}\sum_{t=1}^{12} y_{i,t}$. We estimate equation 3 by OLS resulting in the estimator $\hat{\beta}_{FE} = (\sum_{i=1}^{N}\ddot{\mathbf{X}}'_i\ddot{\mathbf{X}}_i)^{-1}(\sum_{i=1}^{N}\ddot{\mathbf{X}}'_i\ddot{\mathbf{Y}}_i)$, where $\ddot{\mathbf{X}}_i$ is a (12×2) matrix consisting of $\ddot{\mathbf{x}}_{i,t}$s and $\ddot{\mathbf{Y}}_i$ is a (12×1) vector of $\ddot{y}_{i,t}s$ both stacked over time. For the estimator $\hat{\beta}_{FE}$ to be consistent, we need strict exogeneity of the input factors with respect to the productivity shocks ($u_{i,t}$) and a rank condition. The first condition is FE.1: $E[u_{i,t}|\mathbf{x}_{i,1}, \mathbf{x}_{i,2}, ..., \mathbf{x}_{i,12}, c_i] = 0 \ \forall \ t$. The second condition is FE.2: $rank(E[\ddot{\mathbf{X}}'_i\ddot{\mathbf{X}}_i]) = 2$, i.e. the input factors vary within firms over time and the within-transformed log of capital and log of employment cannot be perfectly collinear. FE.2 is satisfied in our setting as illustrated by figure 1. FE.1 is quite a strong assumption, as it rules out any feedback effects from previous (or future) productivity shocks. For instance, if productivity shocks are persistent to some degree, one might expect firms to respond to past positive (negative) shocks by increasing (decreasing) their capital and employment in later periods. FE.1 does not allow for such a feedback mechanism.

If we instead remove the time-invariant component of firm productivity (c_i) by taking first differences, we get a FD model, which relies on a slightly weaker form of exogeneity.

The FD model is estimated by the equation below:

$$\Delta y_{i,t} = \Delta \mathbf{x}_{i,t}\beta + \Delta u_{i,t} \quad \forall\ t = 2, 3, .., 12 \tag{4}$$

For the estimator $\hat{\beta}_{FD} = (\sum_{i=1}^{N}\sum_{t=2}^{T}\Delta\mathbf{x}'_{i,t}\Delta\mathbf{x}_{i,t})^{-1}(\sum_{i=1}^{N}\sum_{t=2}^{T}\Delta\mathbf{x}'_{i,t}\Delta y_{i,t})$ to be consistent we require FD.1: $E[u_{i,t}|\mathbf{x}_{i,t-1}, \mathbf{x}_{i,t}, \mathbf{x}_{i,t+1}, c_i] = 0\ \forall\ t$ and FD.2: $rank(\sum_{t=2}^{T} E[\Delta\mathbf{x}'_{i,t}\Delta\mathbf{x}_{i,t}]) = 2$. FD.2 is unproblematic in our setting. FD.1 rules out that firms adjust their capital- or labor input in response to a shock from the period just before (or after). Whether this assumption is likely to hold depends on how easily firms can adjust their capital and hire/fire employees. In economics, we often think of capital as being fixed in the short run; e.g. it might take several years to build a new factory. Labor is often thought of as being more easily adjusted. However, institutional factors that make it hard to fire (and thus also more risky to hire) can also make firms unable/unwilling to adjust labor input in the short run. France is known as a country in which it is particularly difficult to fire employees. Thus, in our empirical setting, firms may respond to past productivity shocks with *a lag*, such that FE.1 is violated without necessarily the same being true for FD.1.

4 Empirical Analysis

4.1 Exogenerity conditions

As discussed in section 3, the French empirical setting leads us to separately estimate a FE and a FD model, to eliminate any time-invariant firm specific productivity components. As strict exogenerity (FE.1) is required for consistent estimation of the FE model parameters, we start the analysis by testing an implication of this assumption. Specifically, we estimate the following equation

$$\begin{aligned} \ddot{y}_{i,t} &= \ddot{\mathbf{x}}_{i,t}\beta + \ddot{\mathbf{x}}_{i,t+1}\gamma + \ddot{\epsilon}_{it} \quad \forall\ t \neq 12 \\ \Leftrightarrow \ddot{y}_{i,t} &= \underbrace{\ddot{\mathbf{z}}_{i,t}}_{(1\times 4)}\theta + \ddot{u}_{it} \quad \forall\ t \neq 12 \end{aligned} \tag{5}$$

, where $\ddot{\mathbf{z}}_{i,t} = (\ddot{k}_{i,t}\ \ddot{l}_{i,t}\ \ddot{k}_{i,t+1}\ \ddot{l}_{i,t+1})$ and $\theta = (\beta'\ \gamma')' = (\beta_K\ \beta_L\ \gamma_K\ \gamma_L)'$. FE.1 implies that $E[y_{i,t}|\mathbf{X_i}, c_i] = E[y_{i,t}|\mathbf{x}_{i,t}, c_i]$, and hence future values of capital and labor should not affect output, i.e. $\gamma = \mathbf{0}$.

Formally, we test the hypothesis, $H0 : \underbrace{\mathbf{R}}_{(2\times4)}\underbrace{\hat{\theta}}_{(4\times1)} = \underbrace{\mathbf{r}}_{(2\times1)} \Leftrightarrow (\hat{\gamma}_K\ \hat{\gamma}_L)' = (0\ 0)'$, using the following Wald test

$$W := (\mathbf{R}\hat{\theta} - \mathbf{r})'(\mathbf{R}(\hat{\mathbf{V}}/N)\mathbf{R}')^{-1}(\mathbf{R}\hat{\theta} - \mathbf{r}) \tag{6}$$

, where the estimated variance/covariance matrix,

$$\hat{\mathbf{V}} = (\frac{1}{N}\sum_{i}^{N} \ddot{\mathbf{Z}}_i'\ddot{\mathbf{Z}}_i)^{-1}(\frac{1}{N}\sum_{i}^{N} \ddot{\mathbf{Z}}_i'\hat{\ddot{\mathbf{u}}}_i\hat{\ddot{\mathbf{u}}}_i'\ddot{\mathbf{Z}}_i)(\frac{1}{N}\sum_{i}^{N} \ddot{\mathbf{Z}}_i'\ddot{\mathbf{Z}}_i)^{-1}$$

, is robust to heteroskedasticity and any arbitrary form of serial correlation of productivity shocks within firms. $\hat{\ddot{\mathbf{u}}}_i = \ddot{\mathbf{Y}}_i - \ddot{\mathbf{Z}}_i\hat{\theta}$ is a (441×1) vector of residuals. $\hat{\mathbf{V}}$ is consistent under FE.1 and FE.2. Under H0, we have $\mathbf{R}\theta = \mathbf{r}$, thus we can rewrite $(\mathbf{R}\hat{\theta} - \mathbf{r}) = \mathbf{R}(\hat{\theta} - \theta)$. FE.1-2 then implies $\mathbf{R}\sqrt{N}(\hat{\theta} - \theta) \overset{d}{=} N(\mathbf{0}, \mathbf{RVR}')$. Hence the Wald test-statistic is asymptotically χ^2-distributed with two degrees of freedom under H0.

Table 1 shows that $\hat{\gamma}_K, \hat{\gamma}_L$ are jointly significant with a test-statistic $W \approx 44$ and a p-value of 0. Thus, we reject the H0 and conclude that strict exogenerity is likely a too strong assumption. This result suggests that firms respond to productivity shocks by adjusting their labor and capital inputs. We obtain the same conclusion when constructing a robust version of the Hausman test of strict exogenerity. In this case, we reject the H0 that FE.1-2 and FD.2 are satisfied. The intuition behind the test is that, under H0, $\hat{\beta}_{FE}$ and $\hat{\beta}_{FD}$ have the same probability limit and hence one would expect "similar" coefficient-estimates (Wooldridge, 2010, page 325). We report the Hausman test-statistic and corresponding p-value in table 1.

The first-difference model relies on a weaker exogenerity condition (FD.1), which implies $E[y_{i,t}|\mathbf{x}_{i,t-1}, \mathbf{x}_{i,t}, \mathbf{x}_{i,t+1}, c_i] = E[y_{i,t}|\mathbf{x}_{i,t}, c_i]$. The FD.1 condition restricts production inputs to be orthogonal to productivity shocks in the immediate past, present, and future. Unlike the fixed effects model, firms are allowed to adjust labor and capital inputs in response to a productivity shock in the *long run*, as no restrictions are placed on the relationship between a productivity shock in period t and input factors in period $t + h$, where $h > 1$.To gauge whether this weaker form of exogenerity apply in the empirical setting we estimate the following equation

$$\Delta y_{i,t} = \Delta\mathbf{x}_{i,t}\beta + \mathbf{x}_{i,t}\delta + \Delta u_{i,t} \quad \forall\ t = 2, 3, .., 12 \tag{7}$$

, where $x_{i,t} = (k_{i,t}\ l_{i,t})$ and δ is a (2×1) vector with corresponding coefficients. Under FD.1, $\mathbf{x}_{i,t}$ should have no explanatory power. This leads us to test the hypothesis $H0 : \delta = \mathbf{0}$. We construct a Wald test statistic similar to that of equation 6. From table 1, it is evident that $\hat{\delta}_K$, $\hat{\delta}_L$ are jointly insignificant with a Wald-test statistic of $W \approx 3.4$ and a corresponding p-value of 0.18. Hence, we fail to reject the H0, which suggests that the conditions required for consistency of the FD model parameters are satisfied.

The analysis above provides evidence against a strict exogenerity condition as required for consistency of the FE-estimator but evidence in favor of the weaker exogenerity condition required for consistency of the FD-estimator. These results are in line with labor and

especially capital being fixed in the *short run* but adjustable in the *long run*. Based on these findings, we believe that β_K and β_L can be estimated consistently using first-differences.

4.2 Empirical results

Table 2 reports parameter estimates of $\hat{\beta}_K$ and $\hat{\beta}_L$ for the POLS, FE, and FD models. The estimated coefficients from the FD model in column 3 show that a one percent increase in employment increases output by roughly 0.55 percent ($\hat{\beta}_l = 0.55$), whereas a one percent increase in capital increases output by 0.06 percent ($\hat{\beta}_k = 0.06$).

As described in section 3, constant returns to scale implies that $\beta_K + \beta_L = 1$. The relevant null hypothesis is therefore, $H0 : \mathbf{R}\hat{\beta} = r$, where $\mathbf{R} = (1\ 1)$ is a (1×2) vector and $r = 1$ is a scalar. To test this hypothesis in the FD model, we construct the following Wald test statistic

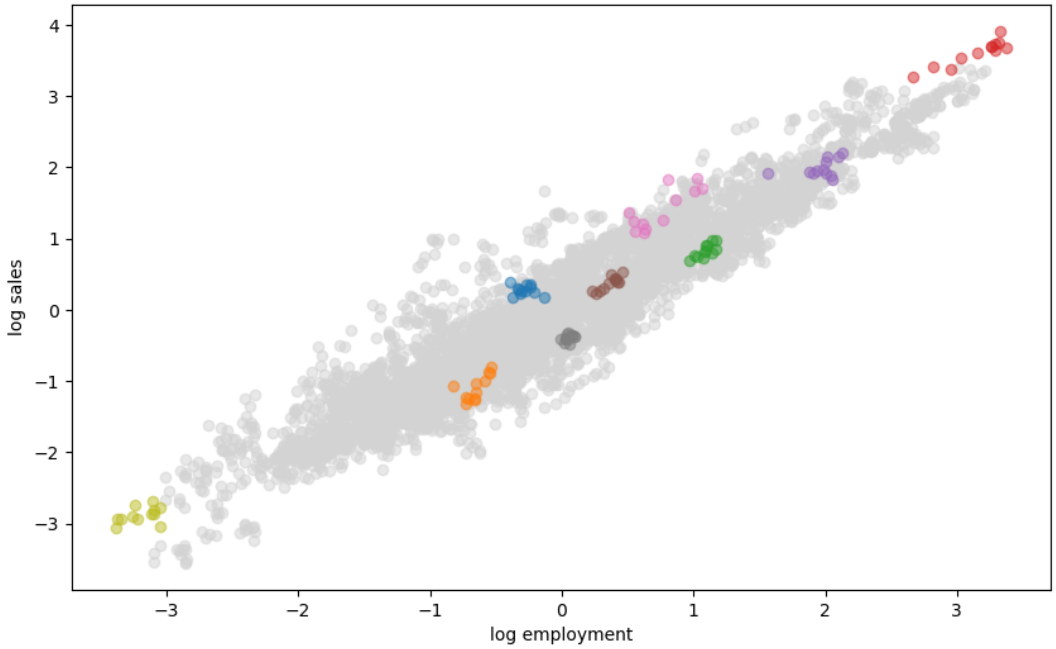
$$W := (\mathbf{R}\hat{\beta}_{FD} - r)'(\mathbf{R}(\hat{\mathbf{V}}/N)\mathbf{R}')^{-1}(\mathbf{R}\hat{\beta}_{FD} - r) \tag{8}$$

, where $\hat{\mathbf{V}}$ is a (2×2) robust variance/covariance matrix. Under H0 and FD.1-2, the Wald test-statistic is χ^2-distributed with one degree of freedom. In table 2, we have reported the Wald test-statistic for the H0 of constant returns to scale. We obtain a test-statistic of $W = 150$ and a corresponding p-value of 0, thereby rejecting the hypothesis that French firms exhibit constant returns to scale.

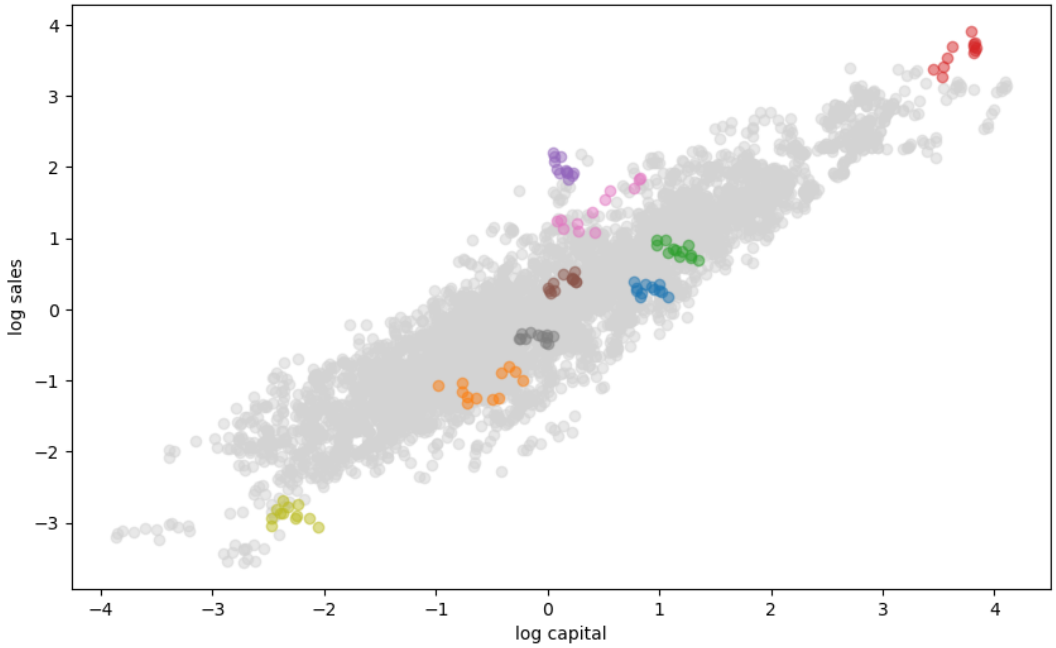
5 Discussion

In this project, we show that constant returns to scale in production might be empirically questionable. Utilizing a panel dataset covering 441 French firms across 12 years and a first-difference model, we find no evidence in favor of constant returns to scale.

The validity of our result hinges on an exogenerity condition that restricts firm behavior; firms are only allowed to adjust production inputs in response to a productivity shock with a lag of *at least* a year. Further, firms are not allowed to adjust input factors if expecting a productivity shock the next year. Capital is often assumed fixed in the *short run*. As argued in section 3, the empirical setting allows for labor to be fixed in the short run, since it is particularly hard to fire (and hence risky to hire) in France. However, it remains unclear from a theoretical point of view how *short* the *short run* really is. Nevertheless, our empirical analysis in section 4 supports the necessary exogenerity condition and hence the implied restrictions on firm behavior.



(a) Labor



(b) Capital

Figure 1: Firm clustering

Note: Each (i, t) observation correspond to a single point i both figures. Colored points show a subset of the unique 441 firms in the data.

TEST	H0	Test statistic	P-value	Distribution under H0	Conclusion
Strict Exogenerity	$\gamma_K = \gamma_L = 0$	44.11	0.00	$\chi^2_{(2)}$	Reject
Hausman	FE.1-2 & FD.2	70.31	0.00	$\chi^2_{(2)}$	Reject
Exogenerity (FD.1)	$\delta_K = \delta_L = 0$	3.41	0.18	$\chi^2_{(2)}$	Don't reject

Note: The conclusion is based on a 5% significance level.

Table 1: Test of exogenerity conditions

	Pooled OLS	FE	FD
Log of capital ($\hat{\beta}_K$)	0.310	0.155	0.063
	(0.0324)	(0.0299)	(0.0232)
Log of labor ($\hat{\beta}_L$)	0.675	0.694	0.549
	(0.0366)	(0.0417)	(0.0292)
Test CRS (H0: $\beta_K + \beta_L = 1$)			
Wald test statistic	1.56	19.40	150.03
P-value	0.212	0.000	0.000
Conclusion	Don't Reject	Reject	Reject

Note: Robust standard errors in parentheses.

Table 2: Results

References

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*, Volume 108. MIT press Cambridge, MA.